



DATA LIFECYCLE

A GENERAL VIEW

The diagram consists of three large, light-orange rectangular boxes arranged horizontally. Each box is divided into two horizontal sections. The top section of each box contains a dark-orange square with a white number (1, 2, or 3). The bottom section contains a text label. The boxes are connected by a dark-orange horizontal bar at the top, which has three vertical dark-orange bars extending downwards from it, separating the three boxes. Above the top bar, there are three horizontal bars of different colors: dark grey, orange, and light grey.

1

Methodologies

2

tools

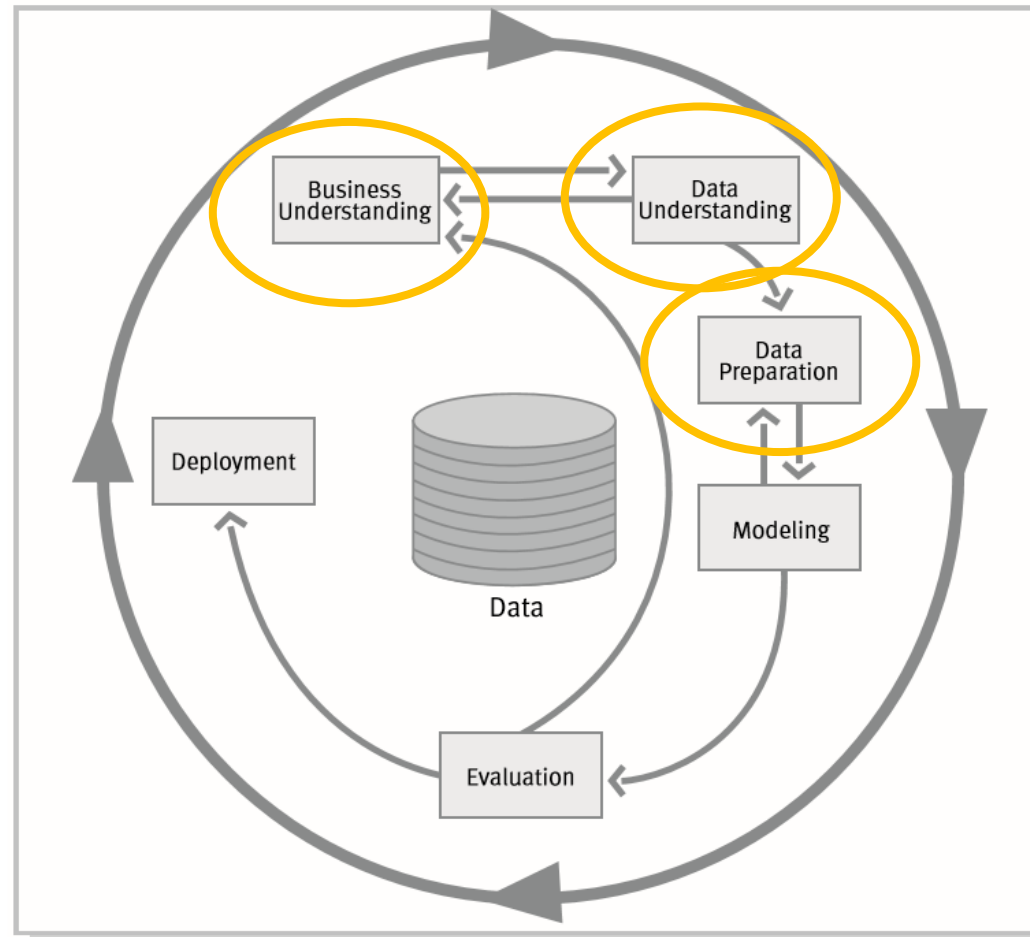
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Phases

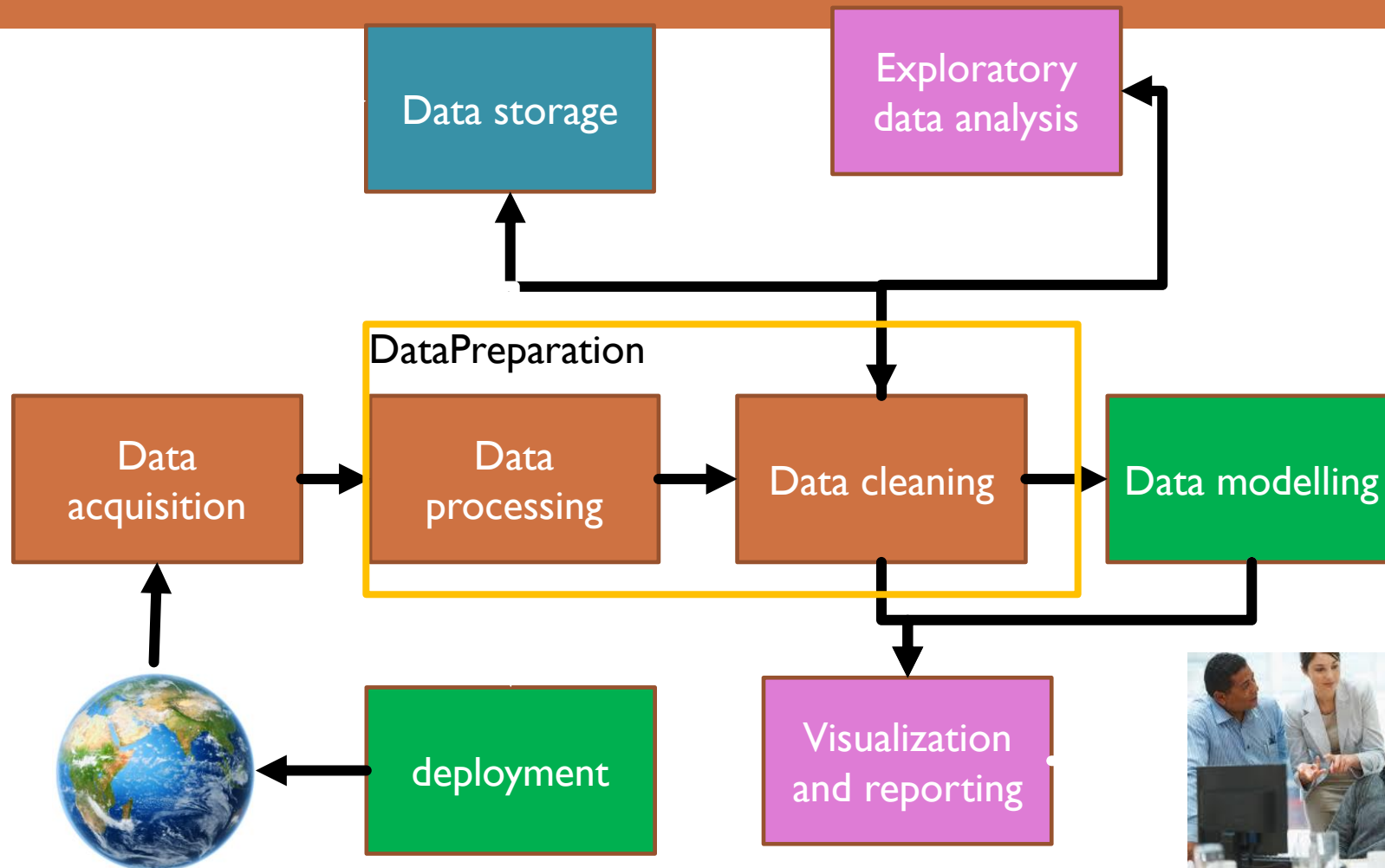
DATA LIFECYCLE

- Set the goal of research
 - What is the question?
 - The computers are useless. They can only give you answers. (Picasso)
- Find the data
 - Where are the datasets I need?
- Data preparation (about 70% of the whole process)
- From raw data to useful data
 - Data exploration
 - Partially covered by the data visualization module
 - Data modeling
 - Not covered in this course
- Presentation and distribution
 - Partially covered by the data visualization module

THE CRISP-DM REFERENCE MODEL OF THE LIFE CYCLE OF ANALYSIS

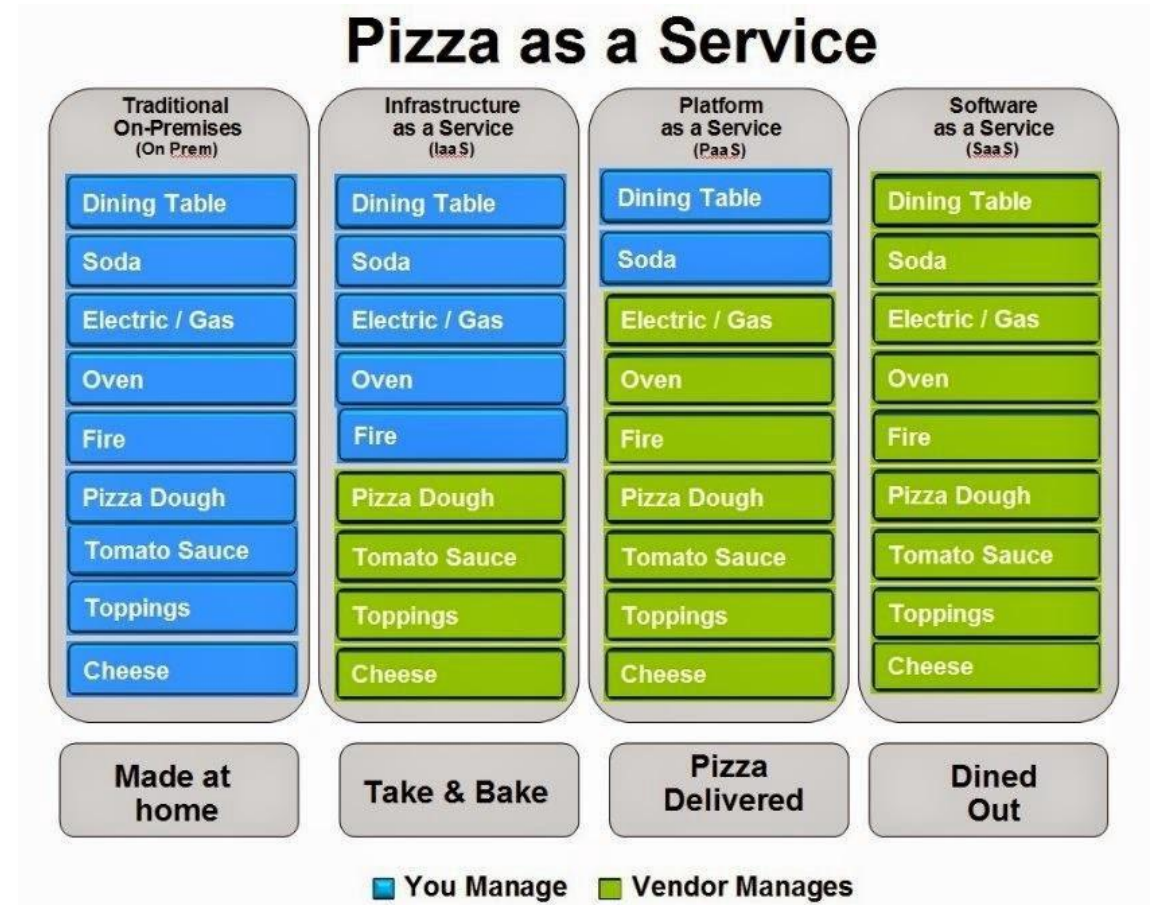


DATA SCIENCE PIPELINE



WHAT TYPE OF TOOLS?

- Each step in the DS process can be performed by means of different tools
 - From do it yourself to SaaS tool
- It is up to you the balance among
 - Easy to use
 - Cost/time
 - Availability
 - Needs
 -



DATA ACQUISITION

- Where can I find my data?
- Internal (I'm the owner)
 - Operational data
 - Clients
 - Logs
 - IoT
- External (I'm not the owner)
 - Social
 - Web
 - Open data
- Internal data
 - Who knows where the data is?
 - What about the access right?
- External
 - How much does it cost?
 - Free / Freemium
 - See open corporates (<https://opencorporates.com/registers>)
 - Payment
 - Cerved group (<https://buydirect.cerved.com/it/homepage>)
- The gray area
- Pay attention to:
 - Format
 - Quality

FIND DATA: DATA ACQUISITION TECHNIQUES

- Once a data set is identified, how can I get it?
- Download
 - Kaggle.com
- Queries
 - SQL endpoint
- API
- Scraper

API

- Application program interface
- It represents the way in which a software component can interact with another one
- API rest

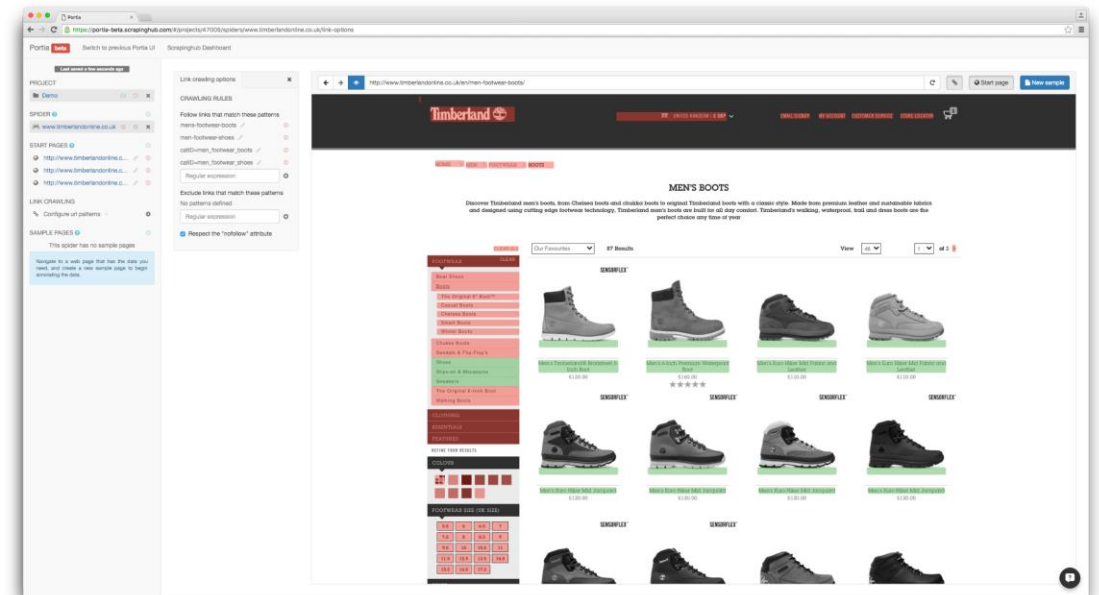
Standard CRUD Operations

This table lists the standard create, read, update, and delete (CRUD) operations for a business object service.

Operation	Method Name	Description
create Operation	<code>createServiceDataObjectName</code>	Creates a service data object and its descendants.
delete Operation	<code>deleteServiceDataObjectName</code>	Deletes a business object.
get Operation	<code>getServiceDataObjectName</code>	Retrieves a single business object by the primary key.
find Operation	<code>findServiceDataObjectName</code>	Finds and returns a list of business objects that meet the specified search criteria.
find by additional predefined search criteria Operation	<code>findServiceDataObjectNameSearchCriteriaName</code>	Finds and returns a list of business objects that meet the specified search criteria and the additional predefined search criteria.
update Operation	<code>updateServiceDataObjectName</code>	Updates a business object.
merge Operation	<code>mergeServiceDataObjectName</code>	Updates a business object, if it exists. Otherwise creates a new business object.
process Operation	<code>processServiceDataObjectName</code>	Performs create, update, delete, or merge operation on a list of business objects. The specified operation is applied to all the objects in a given list.
process change summary Operation	<code>processCSServiceDataObjectName</code>	Performs create, update, or delete operations on a list of business objects. Allows you to specify different operations for different objects.

WEB SCRAPING

- Set of technologies able to browse single html page to collect data

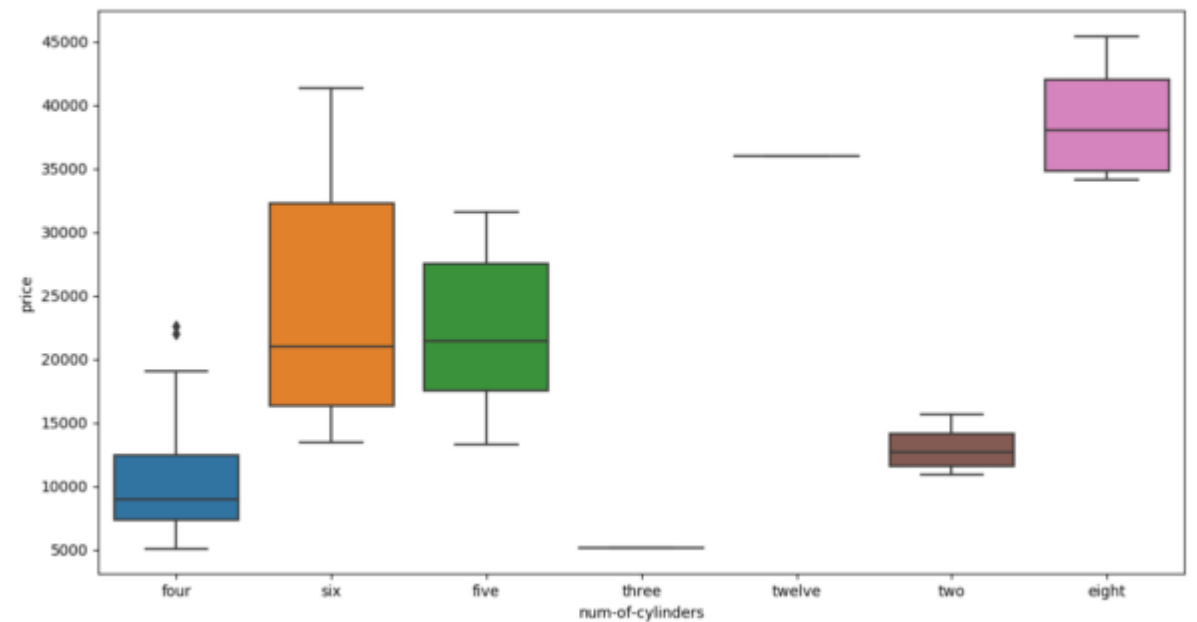
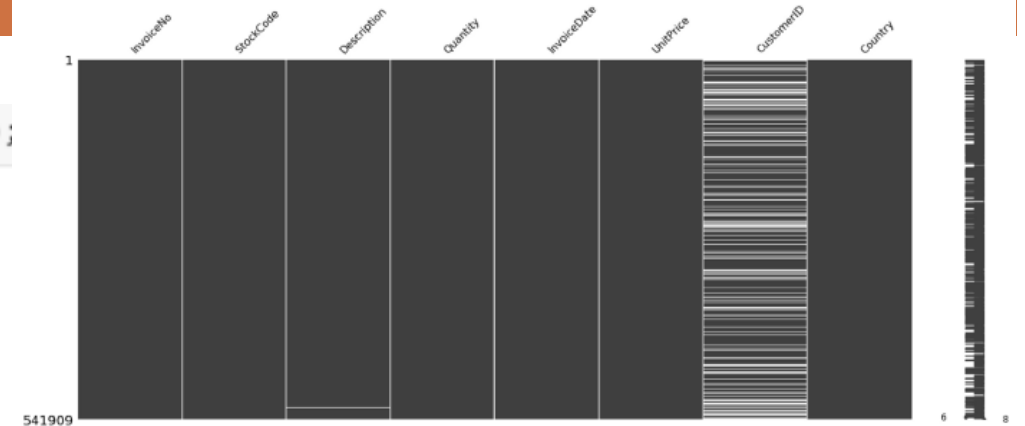
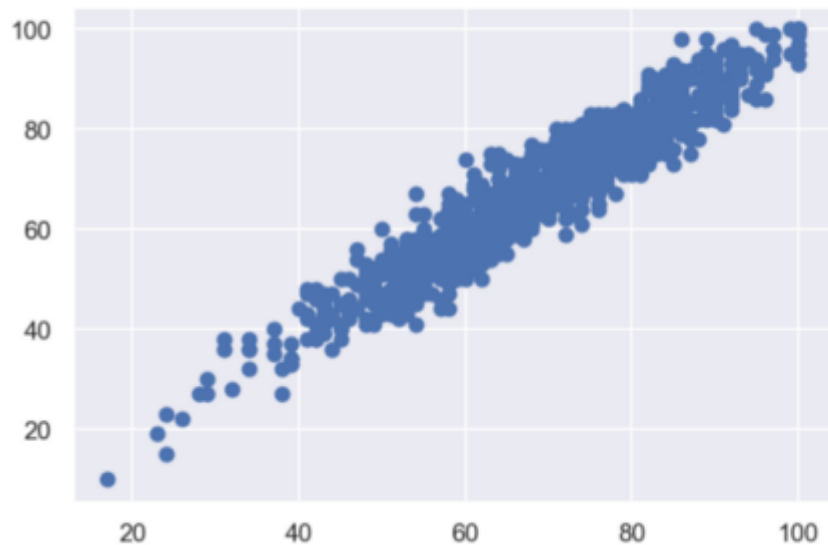


DATA STORAGE

- How data is modeled (relational, json, excel ...)
- How the data is described (Surname, Name NameSurname)
- How to store the integrated data set (virtual, consolidate)

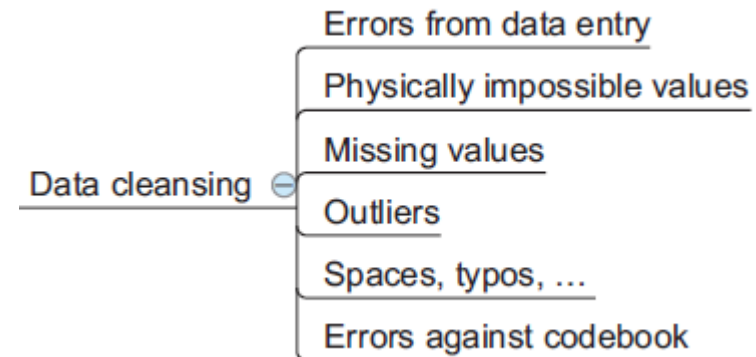
DATA UNDERSTANDING: EXPLORATORY DATA ANALYSIS

```
plt.scatter(Performance_df['Reading_Score'], Performance_df['Writing_Score']);
```



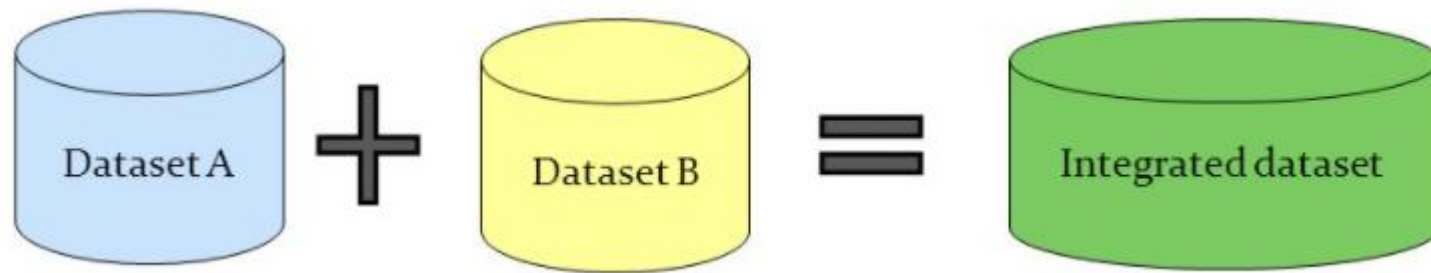
DATA PREPARATION: CLEANSING

- The world is dirty!
 - And the data too!
- Data quality defined as “fit for use”
- Complex if you want to use the same data set for different goals
- Quality dimensions
 - Precision
 - Completeness
 - Consistency
 - Time-related dimensions
- Tools, techniques, coding



DATA PREPARATION: INTEGRATION

- There are usually more than one dataset in a data science project
- How to integrate the data?
- How to merge values (Andrea is a male, Andrea is a female, different salaries for the same person)



- A=sales db, B=
- associate to each product its social reputation

DATA MANAGEMENT

- The entire data science process can be executed multiple times,
- The same dataset can be used in different data science processes
- How to manage?
 - Data lake approach
 - Master data management



EXERCISE



EXAMPLE I

- How to advertise students that there is a new master's degree?
- Among others ... students from the same university
- Who and how to contact them?
- OBJECTIVE:
 - Find the name, surname and email address of all potential "local" candidates for the new master
 - What does "potential" mean

EXAMPLE 2

- For the new website of the department, we need to organize research laboratories (about 18) in max 5-6 "areas" with a "catchy" name
- How to group laboratories in 5/6 areas?
- What data do we need?